Zhijie Lyu

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Education

Stony Brook University – SUNY January 2020 – May 2026
PhD student in Mechanical Engineering with a minor in Computer Science *Stony Brook, New York, USA*

Stony Brook University – SUNY August 2018 – January 2020
Master's student in Mechanical Engineering, Manufacturing Direction *Stony Brook, New York, USA*

Southwest Jiaotong University August 2014 – May 2018
Bachelor in Mechanical Engineering, Mechatronic Direction *Sichuan, Chengdu, PRC*

Experience

Coursework August 2018 – December 2021
GPA during master's: 3.93; GPA during Ph.D.: 3.80

- Topics: Robotics, Computer-Aided Design, Data Driven Design, Analysis of Algorithms, Computational Geometry

Ph.D. Research January 2020 – Present
Focus: Data-Driven Design, Data Structures, and Algebraic Graph Theory

- **Journal Publication Record**: 1 for Data-Driven Synthesis Pipeline (Selected as featured article for ASME JMD); 1 for Mechanism Dataset; 3 for Efficient Geometric Constraint Solving; 1 for Mechanism Cognate Generation, as second author.
- **Journal Review Record**: multiple manuscripts from ASME-JMD, ASME-JMR and IEEE-THMS
- **On-going Research:** Differentiable and parallelized computation framework for CPUs and GPUs.

Teaching January 2020 – Present
Served as a teaching assistant for undergraduate students and high school students

- Worked as TA for four undergraduate courses: Intro to M. Eng., Statics, Dynamics, Numerical Methods, and Mechanism Design
- Guided a high school student to develop a framework for mechanism design; got the work published in MDPI

Academic Presentation August 2022, 2023, 2024, and 2025
Participated in four ASME IDETC/CIE and presented our academic research results.

Software Development (Funded by NSF and Mechanismic Inc.) May 2019 – Present
*Development and Evaluation of MotionGen (**motiongen.io**, a web-based CAD application)*

- **Display**: implementation of customizable shape elements on the canvas using paper.js **(2019 – 2020)**
- **Mechanism Simulation**:
  1. Cooperated on a unified, generalizable, differentiable and efficient planar linkage simulator using approximation **(2022)**
  2. Extended the simulation capability using a Pebble Game framework for gears and wheels into the linkage system with precise constraint system. **(2024 – 2025)**
- **Mechanism Synthesis**:
  1. Deployed the G-manifold algebraic fitting method for motion synthesis **(2022)**
  2. Deployed a backend server using Google Cloud and a neural network pipeline **(2023)**
  3. Designed a compact data structure that reduced the storage of the same information data to $1/100$ of its original in the code base, and reduced the query time with the use of customized topology dictionary **(2024)**
  4. Generated a massive mechanism dataset for these mechanisms **(2024)**
  5. Added differential evolution method to the synthesis framework **(2025)**
- **On-going Development (2025-2026)**:
  1. Vectorization (matrix-formatted parallel programming) of the simulator for MotionGen, intended for real-time synthesis
  2. Differentiable physics intended for synthesis fine-tuning
  3. Multibody dynamics, intended for real-time rendering
- **Impact**: The number of registered users is approximately $100,000$ as of November 2025, since it had a registration system in 2020. A sharp increase in user registration occurred in August 2023 when a popular mechanism design channel Engineezy used our application to design a mechanism prototype.

Technical Skills, Language Skills, and Interests

Programming Languages: JavaScript, Python, MatLab scripts, C++ (OOP)
Writing: LaTeX, Office
Languages: Cantonese (native), Mandarin Chinese (native), English (fluent)
Interests: Prompt Engineering, Software Development, Applied Math Science, High Performance Computation (HPC), Document Translation, Speedrunning in video games

Selected Publications (The first three have been implemented in MotionGen 2024)

A Unified Real-Time Motion Generation Algorithm for Approximate Position Analysis of Planar N-Bar Mechanisms - ASME Journal of Mechanical Design, June 2024, 146(6): 063302 (12 pages)
Authors: ***Zhijie Lyu***, Wei Liao, Anurag Purwar

- Extended the prismatic joint approximation trick from specific cases in textbook to arbitrary combination cases;
- Defined a standardized matrix-set data structure for kinematically simulated samples, and their structural complexity analysis;
- *This documents the core of the 2022-2025 MotionGen simulator.*

Deep Learning Conceptual Design of Sit-to-Stand Parallel Motion Six-Bar Mechanisms - ASME Journal of Mechanical Design, July 2024, 147(1): 1-18
Authors: ***Zhijie Lyu***, Anurag Purwar

- Improved an atlas-based and data-driven approach for planar linkage path synthesis during data preprocessing;
- Used a Variational Auto-Encoder (VAE) along with three metrics to rank solution candidates retrieved from the dataset.
- *The work is selected as a featured article on ASME Journal of Mechanical Design by editor Faez Ahmed from MIT.*

A Dataset of 3M Single-DOF Planar 4-, 6-, and 8-bar Linkage Mechanisms with Open and Closed Coupler Curves for Machine Learning-Driven Path Synthesis, ASME Journal of Mechanical Design (Special Issue for IDETC-CIE 2024)
Authors: Anar Nurizada, Rohit Dhaipule, ***Zhijie Lyu***, Anurag Purwar

- A dataset of three million samples of R-joint and/or P-joint mechanisms is created with the simulation package of MotionGen;
- *This dataset, together with a compact kinematic structure library, is implemented onto MotionGen using Google Cloud API with the synthesis method presented in the second publication.*

Point-based Models: A Unified Approach for Geometric Constraint Analysis of Planar N-Bar Mechanisms with Rotary, Sliding, and Rolling Joints, ASME Journal of Mechanisms and Robotics (Special Issue for IDETC-CIE 2024)
Authors: ***Zhijie Lyu***, Anurag Purwar

- Developed the methodology and data structure to extend the constraint graph approach from two types of constraints (distance, angle) to three types (angle-angle); This is intended for the implementation of wheels/gears into the linkage system.
- *This is theory part that says wheels can be represented as linkages, so we can analyze all the elements like analyzing a truss structure.*

Construction, Reduction, and Decomposition of Planar Mechanisms via an Extended Pebble Game Framework, ASME Journal of Mechanisms and Robotics, December 2025, 17(12): 121009 (15 pages)
Authors: ***Zhijie Lyu***, Anurag Purwar

- Developed a modified pebble game framework to extend the constraint graph approach from two types of constraints (distance, angle) to three types (angle-angle); The framework analyzes the planar geared linkage system in polynomial time.
- The package has been completed in a Python-based prototype.
- *This is a polynomial-time cost algorithm for general mobility analysis, but is expanded from linkages to gears and linkages.*

Path Generative Model Based on Conditional β-Variational Auto Encoder for Four-Bar Mechanism Design, ASME Journal of Mechanisms and Robotics, June 2025, 17(6): 061004
Authors: Anar Nurizada, ***Zhijie Lyu***, Anurag Purwar

- The paper itself is mostly led by Anar Nurizada. The research goal is given some paths, the neural network outputs an initial guess on the mechanism configurations.
- I made an automated script on generating the four-bar mechanism cognates for any given four-bar linkage, and contributed to the text related to this part.

(Accepted) Data-Driven Design using Differentiable Physics: Sampling Density for Reliable Initialization into Basin of Attraction, ASME IDETC/CIE, 2026
Authors: ***Zhijie Lyu***, Suhas Gangireddy, Anurag Purwar

- Integrated machine learning with the customized differentiable simulator to generate precise and satisfactory designs.
- Experimented on different topologies to check their respective sampling densities.